Information Fusion 104 (2024) 102172



Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus





QMFND: A quantum multimodal fusion-based fake news detection model for social media

Zhiguo Qu [a,b,1], Yunyi Meng [b], Ghulam Muhammad [c,*], Prayag Tiwari [d,1]

[a] Engineering Research Center of Digital Forensics, Ministry of Education, Nanjing University of Information Science and Technology, Nanjing, 210044, Jiangsu, China
[b] School of Computer Science, Nanjing University of Information Science and Technology, Nanjing, 210044, Jiangsu, China
[c] Department of Computer Engineering, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia
[d] School of Information Technology, Halmstad University, Halmstad, Sweden

ARTICLE INFO

Dataset link: https://github.com/olivia-2333/QMFND_quantum_fake_news_detection

Keywords:
Fake news detection
Multimodal fusion
Social network
Quantum convolutional neural network

ABSTRACT

Fake news is frequently disseminated through social media, which significantly impacts public perception and individual decision-making. Accurate identification of fake news on social media is usually time-consuming, laborious, and difficult. Although the leveraging of machine learning technologies can facilitate automated authenticity checks, the time-sensitive and voluminous nature of the data brings considerable challenge for fake news detection. To address this issue, this paper proposes a quantum multimodal fusion-based model for fake news detection (QMFND). QMFND integrates the extracted images and textual features, and passes them through a proposed quantum convolutional neural network (QCNN) to obtain discriminative results. By testing QMFND on two social media datasets, Gossip and Politifact, it is proved that its detection performance is equal to or even surpasses that of classical models. The effects of various parameters are further investigated. The QCNN not only has good expressibility and entangling capability but also has good robustness against quantum noise. The code is available at

1. Introduction

With the prevalence of social media, individuals prefer to access news from social media networks rather than from traditional news outlets. However, the rise of social media has also been a double-edged sword, offering easily accessible and convenient short news at the risk of propagating fake news. Fake news can deceive the public potentially through fabricated text, images, audio, video, or mismatched headlines and graphics intended to attract attention. Consequently, it can drastically distort the truth and influence individuals' decisions, leading to illegal profits and the manipulation of public events [1]. For example, during the Covid-19 epidemic, online fake news spread lead to incorrect treatments and false vaccine effects. This could lead the public to make wrong medical decisions and harm their health. Due to the fast-spreading nature of social media and people's need for instant news, it becomes essential to quickly detect the authenticity of the news. In these circumstances, rapid and accurate detection of fake news has become a critical issue that urgently needs to be addressed in social media.

Due to billions of users, the number of news released by them on social media exhibits massive and explosive growth, which causes fake news detection on social media to become more and more challenging. News on current social networks is often presented in a multimodal manner, with text and image combinations being the most common form of multimodal information. Compared to unimodal data, multimodal data requires information fusion techniques to process the data, which is more complicated. Information fusion techniques encompass data fusion [2], feature fusion [3], decision fusion [4], and more [5]. They involve integrating and processing data, knowledge, and features from multiple data sources. They can help to provide a more comprehensive, accurate, and reliable representation of information. For instance, in medical diagnosis and treatment, combining different medical images (such as MRI and CT scans of a patient) can result in a more comprehensive and accurate diagnosis.

In view of this, researchers have explored the application of machine learning (ML) techniques, including graph neural networks (GNN) [6], natural language processing (NLP) [7,8], and generative adversarial networks [9], to detect fake news automatically. However,

* Corresponding author.
E-mail addresses: 002359@nuist.edu.cn (Z. Qu), 20211221061@nuist.edu.cn (Y. Meng), ghulam@ksu.edu.sa (G. Muhammad), prayag.tiwari@ieee.org (P. Tiwari).

[1] Zhiguo Qu and Prayag Tiwari contribute equally and share co-first authorship.

https://doi.org/10.1016/j.inffus.2023.102172
Received 30 June 2023; Received in revised form 28 November 2023; Accepted 29 November 2023
Available online 30 November 2023
1566-2535/© 2023 Elsevier B.V. All rights reserved.

these methods still suffers to process ambigous features and fail to quantify level of uncertainity. Many studies have shown that quantum machine learning (QML) based on quantum computing principles [10] can be used to perform ML tasks [11–13] such as pattern matching [11], binary classification [12], support vector machines [13], and others. Quantum convolutional neural network (QCNN), as one of the emerging research branches of QML, has been widely used in multiple application scenarios, such as smart healthcare [14] intelligent transportation [15], etc.

In summary, the main contributions can be presented, as follows.

- A quantum multimodal fusion-based fake news detection (QM FND) model is proposed for social media networks. The model integrates quantum encoding, and quantum convolutional neural networks (QCNNs) to process high-dimensional data processing.
- To achieve lower complexity and better accuracy, multimodal features are encoded into a VQC (variational quantum circuit) through amplitude encoding.
- QCNN is proposed for efficient training of quantum multimodal data. The QCNN circuit not only has excellent expressibility and entangling capability, but also good robustness against quantum noise.
- QMFND achieved detection accuracies of up to 87.9% and 84.6% on the Gossip and Politifact datasets, respectively. Additionally, QMFND can alleviate the barren plateau phenomenon.

The remaining part of this paper is organized as follows. Section 2 reviews the related work on fake news detection and quantum neural networks (QNNs). Section 3 presents the proposed QMFND model which comprises data pre-processing, data encoding and QCNN training. Section 4 provides the experimental results and analysis, datasets, runtime environment, baselines, and performance comparison. Finally, the conclusions are given in Section 5.

2. Related work

2.1. Traditional fake news detection models

Traditional fake news detection primarily includes based on article information, social context, and their combinations with external knowledge.

The detection based on article information includes text-based, image-based, and multimodal analysis. In 2019, Ma et al. [16] improved classification performance by using adversarial training to generate more data. Singhal et al. [17,18] combined visual and textual information using the visual geometry group 19 (VGG-19), bidirectional encoder representations from transformers (BERT), and the generalized autoregressive pretraining method of XLNet. In 2022, Segura [19] used unimodal and multimodal approaches for a more detailed classification of fake news. In the same year, Jayakody and Mohammad [20] proposed a system for fake news detection, employing federated learning and blockchain methods to address resource allocation and privacy concerns. In 2023, Luvembe et al. [21] proposed a method that achieved high accuracy, using dual sentiment features.

In fake news detection based on social context, user trustworthiness and news dissemination are commonly used criteria. For instance, in 2020, Nguyen et al. [22] proposed a novel graphical social context representation and learning framework called FANG to detect fake news. FANG captures social context well and is robust despite limited training data. In 2022, based on Transformer, Raza et al. [23] proposed an effective labeling technique in a fake news detection framework to address the lack of labeled data in training models.

GNNs and attention mechanisms are also used to detect fake news by integrating external information. In 2021, Li et al. [24] constructed a star-shaped knowledge graph for factual evidence and news content, using GNNs to identify fake news. In the same year, Hu et al. [25] presented a novel end-to-end GNN model called CompareNet to compare news with a knowledge base of entities for fake news detection.

Deep learning models, particularly CNNs, GNNs, recurrent neural networks (RNNs), and Transformers, have been widely used for fake news detection. These models can automatically learn text and image data patterns and help identify fake news. Moreover, the emergence of pre-trained language models (such as BERT, GPT, etc.) has changed the field of natural language processing, providing new opportunities for fake news detection.

Despite the increasing diversity and sophistication of fake news detection methods, multimodal fake news detection is still the focus of research. Fake news exists not only in text but also in the dissemination of images, videos, and audio, simultaneously. Therefore, the need for multimodal fake news detection, which involves the simultaneous analysis of various types of media content, is growing. In addition, the enormous computational power of quantum computing has not yet been fully investigated in fake news detection.

2.2. Quantum neural networks

QNNs leverage the properties of quantum computing to enhance the performance of neural networks in some tasks. For example, in 2021, Narottama et al. [26] proposed a reinforcement learning-inspired quantum neural network (RL-QNN) to enhance resource allocation efficiency in wireless communication. In 2023, to capture the complexity and uncertainty of sarcasm and sentiment elements in human language, Tiwari et al. [27] proposed a quantum fuzzy neural network (QFNN) with a multi-task learning capability for multimodal sarcasm and sentiment detection. The algorithm combines the fuzzy system and QNN to enhance the expressiveness of sentiment and sarcastic features and exhibits superior performance compared to various existing algorithms.

Quantum circuits are used as convolutional kernels in QCNNs with promising feature extraction capability that may match or even surpass classical convolutional kernels [28]. As shown in Fig. 1, the quantum circuit of the convolutional kernel consists of many unitary operators. Similar to the classical convolution kernel, the quantum convolution kernel passes through the entire image according to the size of a given receptive field and step size. However, the quantum convolution kernel first encodes the input into a quantum state consisting of a number of qubits. Gate operations are then performed in a VQC containing entangled modules with trainable weights. Finally, measurement is performed on these qubits to obtain output. Also, the pooling layer of the QCNN reduces the circuit dimensionality, so as to lower the number of qubits while preserving as much information as possible from previously learned values.

At present, QCNNs are widely used in various application fields such as image classification [29,30], traffic prediction [15], speech recognition [31] and medical diagnosis [14,32,33]. In 2020, Li et al. [29] proposed a quantum deep convolutional neural network (QDCNN) model for image recognition based on a parameterized quantum circuit. Network complexity analysis indicates that the proposed model offers exponential acceleration compared to classical models. In 2021, Yang et al. [31] addressed privacy preservation issues in speech recognition by using a decentralized feature extraction approach that employs a QCNN. In 2022, Qu et al. [15] proposed a novel algorithm using a quantum graph convolutional network to simultaneously capture the temporal and spatial features of traffic data for traffic congestion prediction. In 2022, Ovalle-Magallanes et al. [14] used quantum computing in coronary artery X-ray angiography to construct a hybrid neural network. This network employs a mixed-transfer learning approach, in which a quantum network enhances the performance of a pre-trained classical network. In 2023, Qu et al. [32] introduced a QNN-based multimodal fusion (QNMF) system for intelligent medical diagnosis. QNMF can process multimodal medical data transmitted by Internet of Things (IoT) devices, fuse data from different modalities,

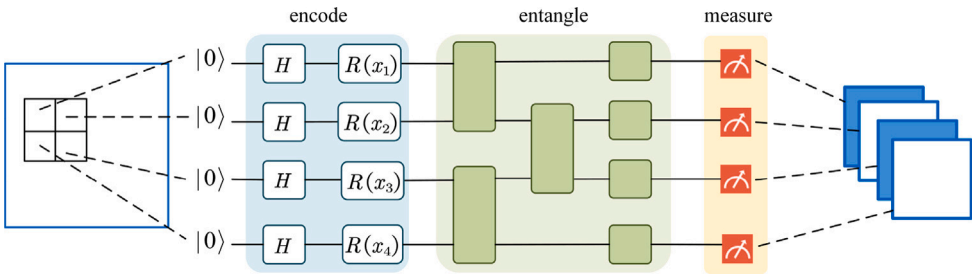


Fig. 1. The quantum circuit of QCNN convolution kernel. $(x_1, x_2, x_3, x_4$ are the parameters. $R(x_i)\,(1 \le i \le 4)$ are the rotation gates. H is the Hadamard gate, and $H = \frac{1}{\sqrt{2}}(|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| - |1\rangle\langle 1|)$.

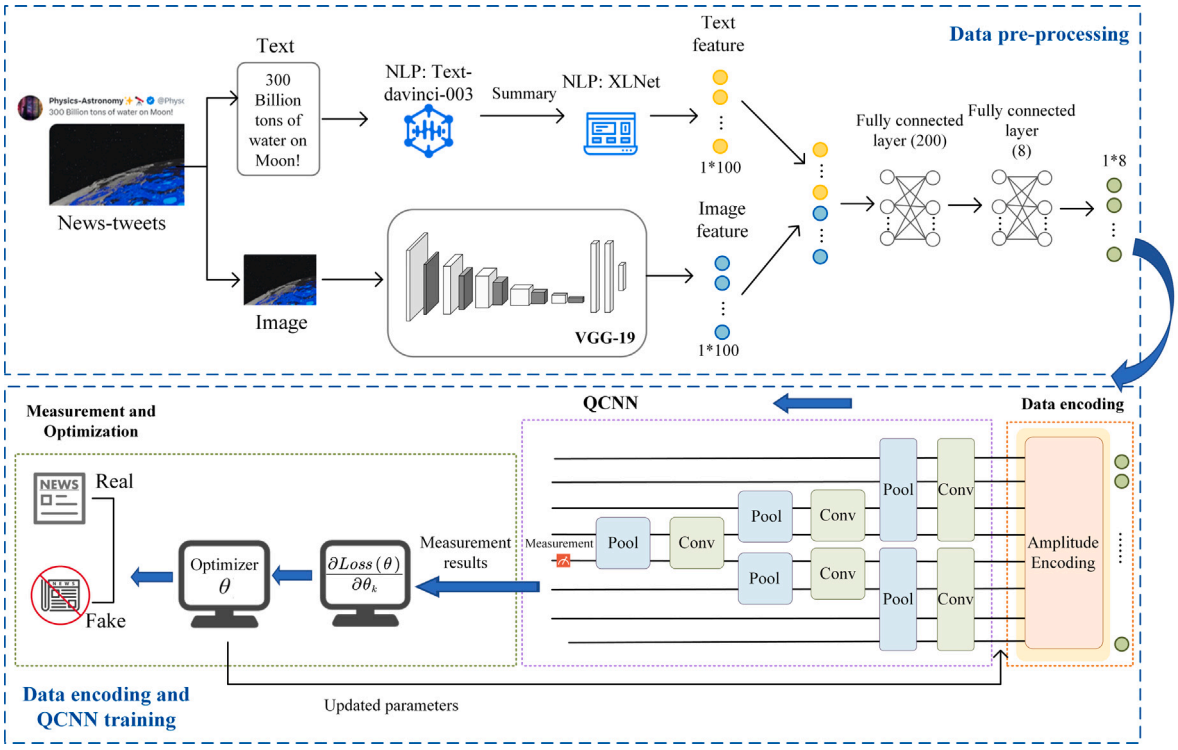


Fig. 2. The flowchart of QMFND.

and enhance the performance of intelligent diagnosis. In 2023, Chen et al. [30] proposed two scale-inspired local feature extraction methods based on QCNN for binary image classification.

The advantages of using QCNNs can be summarized as follows: (1) QCNNs can leverage entanglement, superposition, and interference to process complex tasks. (2) For datasets with specific complex features, QNNs have a classification advantage over classical neural networks [34]. (3) Furthermore, models trained using quantum circuits can exhibit higher performance and superior generalization capabilities because of quantum entanglement [35].

3. The proposed QMFND model

The QMFND model is divided into two main phases. The first phase is the data pre-processing, which separates raw classical data into text and image, and then performs feature fusion. The second phase involves data encoding and QCNN training. The data are fed into the quantum circuit, and the measurement results are used to generate the model prediction results. Then, the output is used to compute the loss function in a classical computer environment, followed by the parameter optimization. The flowchart of QMFND is shown as Fig. 2 and the details are presented as the follow.

3.1. Data pre-processing

3.1.1. Text pre-processing

For news, usually, not all the textual content is strictly relevant to the expressed topic. For example, a news piece may be about the birth of a celebrity's daughter but the text may contain substantial passages describing the celebrity's history.

Quantum natural language processing (QNLP) [36] aims to design and implement NLP models that run on quantum hardware. The release of the Lambeq toolkit by Cambridge Quantum in 2019 marked the first high-level Python library for QNLP. However, our attempts of utilizing Lambeq library revealed that it could only process brief expressions and was time-consuming. QNLP's capability to identify news segments is limited, and the circuit cost is excessively high. Therefore, the classical NLP technique is used to extract text features. The text-davinci-003 model, introduced by OpenAI, is an advanced language model trained to understand and generate human-like text and has been widely recognized for its text-processing capability. Therefore, we uses the API of the OpenAI model to extract the textual content summaries. By calling the API, a non-local model can accurately summarize a relatively lengthy news article by condensing it into a few sentences or phrases. This enhances the training efficiency of the model. Then, news

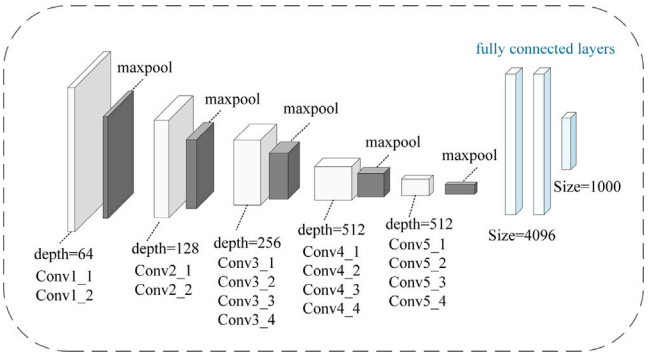


Fig. 3. The structure diagram of VGG-19.

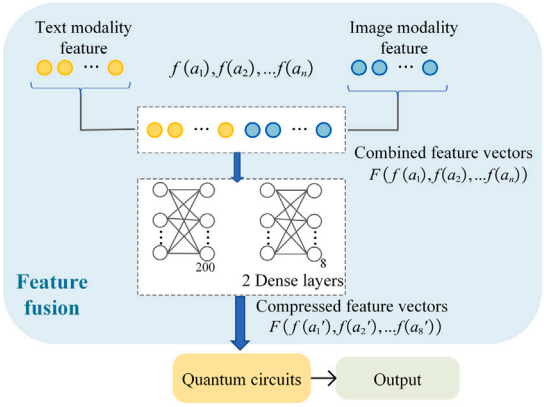


Fig. 4. The flowchart of multimodal feature fusion.

summaries are embedded using a pre-trained XLNet model from the Hugging Face Transformers library to obtain a matrix of eigenvalues. Finally, fully connected layer is used.

3.1.2. Image pre-processing

For news, image information usually plays a vital role in conveying its theme. A pre-trained VGG-19 model is used to extract the image features. VGG-19 is a deep CNN architecture initially proposed by a research team at the University of Oxford in 2014 [37]. As depicted in Fig. 3, VGG-19's architecture comprises 16 convolutional layers and three fully connected layers. The number of convolutional layers in the five VGG blocks is (2, 2, 4, 4, 4), along with three fully connected layers, resulting in 19 parameter layers. VGG-19 is used in the image pre-processing phase because it is typically pre-trained using large-scale image datasets. Thus, its weights already contain a vast amount of image knowledge. This makes VGG-19 highly versatile for various computer vision tasks. Additionally, the VGG-19 network is suitable for extracting rich feature representations. These factors make VGG-19 one of the most popular deep CNN models currently in use.

Considering the limitations of quantum resources, the dimension of image feature vectors is compressed and flattened to 100.

3.1.3. Feature fusion

Information fusion techniques are adept at managing multimodal data. Based on different fusion times, information fusion can be categorized into early, mid, and late fusion. An early feature fusion technique is used, where multiple layers of features are fused at the input layer. Compared with mid fusion and late fusion, early fusion can better utilize complementary information among various modalities, and the extracted features can be effectively used for training the QCNN.

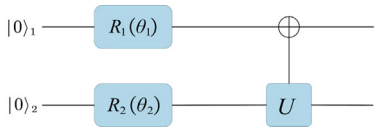


Fig. 5. The quantum circuit that encodes a four-dimensional vector on two qubits.

Let x_i represents the ith data source, and $f(x_i)$ denotes the extracted features from image or text modalities. QMFND concatenates the extracted text and image features, denoted as $F\left(f\left(x_1\right), f\left(x_2\right), \ldots, f\left(x_n\right)\right)$. Through two fully connected layers with the sizes of 200 and 8, the compressed fused multimodal features, denoted as $F\left(f\left(x_1'\right), f\left(x_2'\right), \ldots, f\left(x_8'\right)\right)$, were obtained. A flowchart of multimodal feature fusion is illustrated in Fig. 4. The extracted fused multimodal information will be used at the next encoding step.

3.2. Data encoding and QCNN training

3.2.1. Data encoding

The processed classical data cannot be used directly for training the quantum circuit. Consequently, classical input data must be encoded into a quantum state for further processing in VQC. Given the constraints of quantum resources, circuit simplification costs must be considered in experiments. Reducing the number of qubits is one of the widely accepted approaches. First, amplitude encoding is used to encode classical data into quantum states. It is an effective method for reducing the number of qubits. Specifically, $x = \left(x_1, x_2 \ldots x_N\right)$ represents a normalized vector. Then, the vector can be represented by using the amplitude encoding as follows:

$$|x\rangle = \sum_{i}^{N} x_i |i\rangle, i = 1, 2, \ldots, N. \tag{1}$$

For example, a 4-dimensional classical vector $x = (\alpha, \beta, \gamma, \eta)$ needs to be encoded into the VQC and the initial quantum state is $|00\rangle_{12}$. Applying the rotation gate $R_1\left(\theta_1\right)$ on the first qubit, we get

$$|0\rangle_1 \rightarrow \left(\sqrt{|\alpha|^2 + |\beta|^2}\right)|0\rangle + \left(\sqrt{|\gamma|^2 + |\eta|^2}\right)|1\rangle. \tag{2}$$

Then, the second qubit is also rotated through a rotation gate $R_2\left(\theta_2\right)$:

$$|0\rangle_2 \rightarrow \frac{\alpha|0\rangle + \beta|1\rangle}{\sqrt{|\alpha|^2 + |\beta|^2}}. \tag{3}$$

It can be obtained that:

$$\begin{aligned} |00\rangle_{12} \rightarrow & \left(\sqrt{|\alpha|^2 + |\beta|^2}\right)|0\rangle \frac{\alpha|0\rangle + \beta|1\rangle}{\sqrt{|\alpha|^2 + |\beta|^2}} \\ & + \left(\sqrt{|\gamma|^2 + |\eta|^2}\right)|1\rangle \frac{\alpha|0\rangle + \beta|1\rangle}{\sqrt{|\alpha|^2 + |\beta|^2}}. \end{aligned} \tag{4}$$

Let us assume that there exists a unitary gate U satisfying with:

$$U\left(\frac{\alpha|0\rangle + \beta|1\rangle}{\sqrt{|\alpha|^2 + |\beta|^2}}\right) = \frac{\gamma|0\rangle + \eta|1\rangle}{\sqrt{|\gamma|^2 + |\eta|^2}}. \tag{5}$$

Subsequently, the circuit passes through one more controlled U-gate with the control bit as the first bit. The circuit for amplitude encoding is illustrated in Fig. 5.

After amplitude encoding, it can be obtained that x is encoded into a quantum circuit of the form:

$$\left(\sqrt{|\alpha|^2 + |\beta|^2}\right)|0\rangle \frac{\alpha|0\rangle + \beta|1\rangle}{\sqrt{|\alpha|^2 + |\beta|^2}} + \left(\sqrt{|\gamma|^2 + |\eta|^2}\right)|1\rangle \frac{\gamma|0\rangle + \eta|1\rangle}{\sqrt{|\gamma|^2 + |\eta|^2}}$$

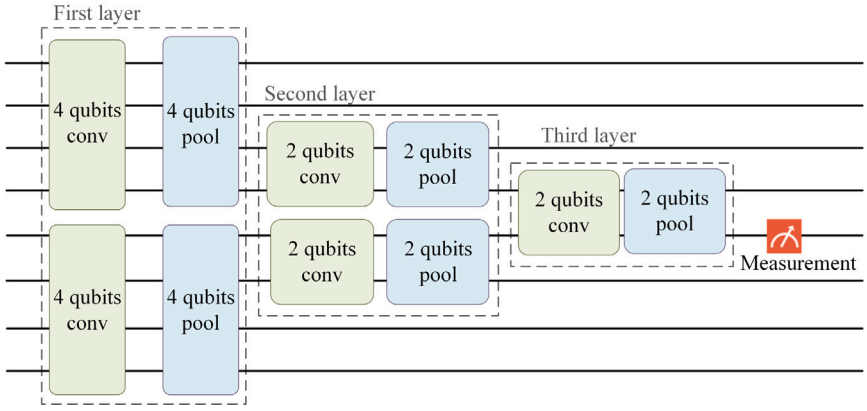


Fig. 6. The QCNN circuit in QMFND (conv represents the convolutional layer and pool represents the pooling layer).

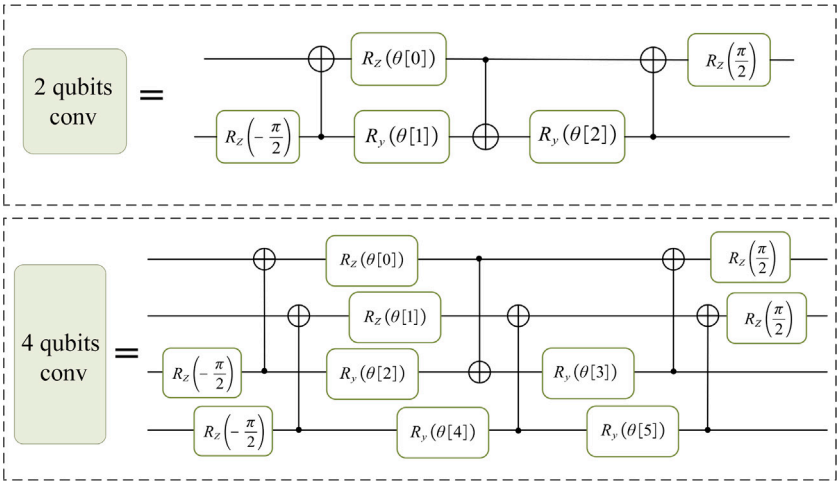


Fig. 7. The quantum circuit of the convolutional layer ($\theta[i]$ is the ith parameter of θ).

$$= \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \eta|11\rangle. \quad (6)$$

It can be seen that only $\lceil \log_2 n \rceil$ qubits are required to represent an n-dimensional vector. In QMFND, eight qubits are used for feature representation.

3.2.2. Quantum circuit of the QCNN

The next step is to construct the QCNN circuit. In this paper, a QCNN is proposed for efficient training of quantum data. The QCNN circuit constructed in this paper is divided into three layers, as shown in Fig. 6. The first layer performs quantum convolution and pooling operations on the first four and last four qubits, respectively. The second layer performs convolution and pooling operations on the third and fourth, and fifth and sixth qubits, respectively. The third layer performs convolution and pooling operations on the fourth and fifth qubits. The quantum measurements are performed on the fifth qubit to obtain the measurement results. The circuits of convolutional and pooling layers are shown in Figs. 7 and 8, respectively. $R_y(\theta)$ and $R_z(\theta)$ are the rotation gates: $R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ -\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$, $R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$. By running quantum convolution and pooling operations in the three layers, the network can extract discriminative features from the input qubits that help detect fake news. The QCNN makes full use of entanglement operations and rotation gates to make the circuits have good expressibility and entangling capability. This helps to represent the correlation between fused features more efficiently so that QMFND can obtain better detection performance.

QCNN discards certain qubits in the circuit, and the discarded qubits are no longer involved in operations and measurements. This successfully compresses the dimensionality of the data and reduces the cost of subsequent circuits.

3.2.3. Measurement and optimization

Multiple measurements are required to improve the performance. The predicted results are the expectation of qubits obtained through several measurements. Cross entropy loss function is used to assess model loss. In addition, we implemented constrained optimization by linear approximation (COBYLA), which is gradient-free. COBYLA ensures efficient optimization of the model parameters, to enhance the optimal performance and reliability. A flowchart of the measurement and optimization is shown as Fig. 9, and the specific steps are presented as below.

Let assume classical inputs are $\{x_i, y_i\}$. x_i is the ith classical data and y_i is the ith label. The real news is labeled as 1 and the fake news is labeled as 0. Set the initial states of the quantum circuit as $|0\rangle^{\otimes n}$ where n is the number of qubits. After the encoding layer, $|\psi\rangle_i$ is obtained.

$$|\psi\rangle_i = U_e|x_i\rangle, \quad (7)$$

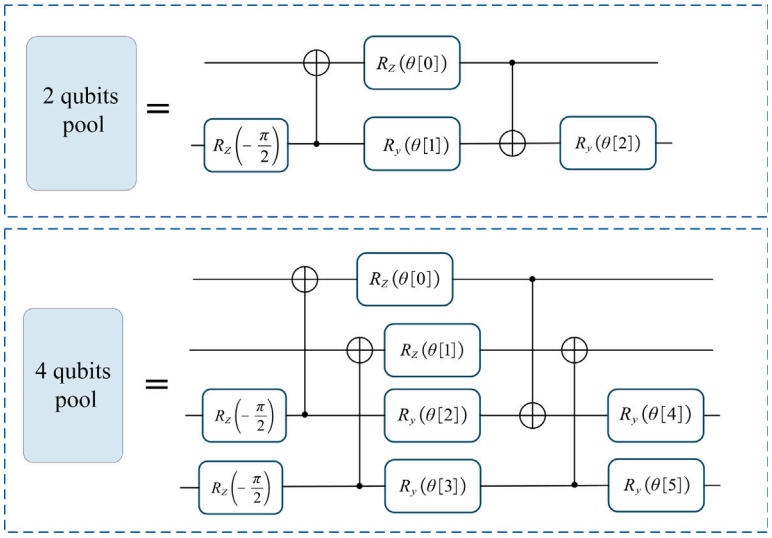


Fig. 8. The quantum circuit of the pooling layer.

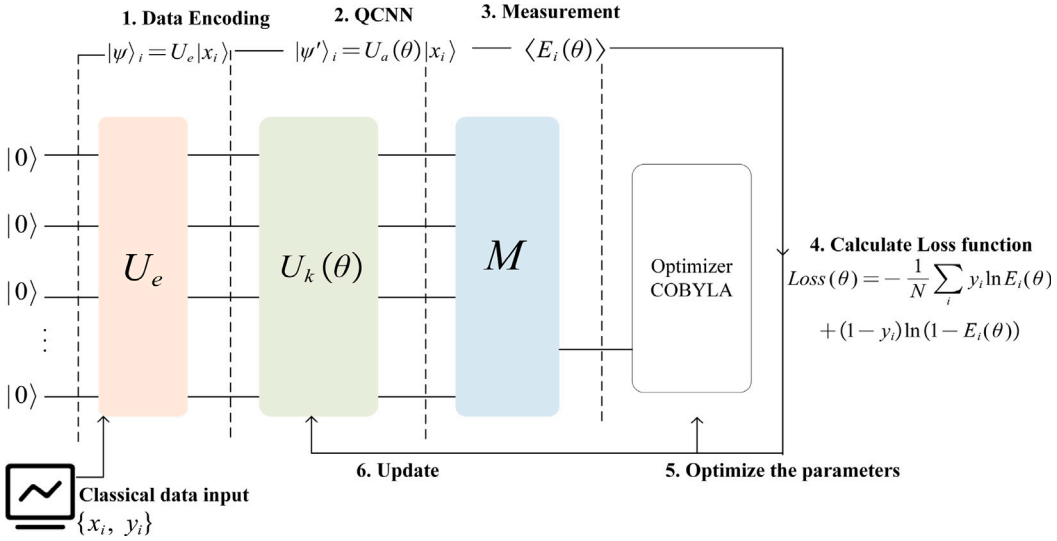


Fig. 9. The flowchart of measurement and optimization.

where U_e denotes the unitary operator of the encoding layer (without parameters). The unitary operator of the QCNN circuit is set as $U_a(\theta)$:

$$U_a(\theta) = \prod_k U_k\left(\theta_k\right). \tag{8}$$

where θ is the parameter to be optimized, θ_k is the kth θ, and U_k is the unitary operators associated with θ_k. After the QCNN layer, the state becomes

$$\left|\psi'\right\rangle_i = U_a(\theta)|\psi\rangle_i, \tag{9}$$

which is measured by using the measurement operator M. $E_i(\theta)$ is the expectation of the predicted values.

$$M = I^{\otimes n-1} \otimes Z, E_i(\theta) = \left\langle\psi'|M|\psi'\right\rangle_i. \tag{10}$$

Then, using cross entropy as the loss function, the equation is given as follows:

$$Loss(\theta) = -\frac{1}{N}\sum_i y_i \ln E_i(\theta) + \left(1-y_i\right)\ln\left(1-E_i(\theta)\right). \tag{11}$$

Here, N is the number of samples in a batch. For the parameter k to be optimized, the gradient of θ_k is

$$\frac{\partial Loss(\theta)}{\partial\theta_k} = -\frac{1}{N}\sum_i \frac{y_i}{E_i(\theta)}\frac{\partial E_i(\theta)}{\partial\theta_k} - \frac{1-y_i}{1-E_i(\theta)}\frac{\partial E_i(\theta)}{\partial\theta_k} \tag{12}$$

It is not difficult to determine whether only the partial derivative of E_i with respect to θ_k is required. Here, E_i is denoted by E. Taking the partial derivative of E with respect to θ_k, we can obtain

$$\begin{aligned}\frac{\partial E(\theta)}{\partial\theta_k} &= \frac{\partial\langle\psi'|M|\psi'\rangle}{\partial\theta_k} = \frac{\partial\left\langle\psi\left|U_a^\dagger M U_a\right|\psi\right\rangle}{\partial\theta_k} \\ &= \left\langle\psi\left|U_a^\dagger M\frac{\partial U_a}{\partial\theta_k}\right|\psi\right\rangle + H.c.\end{aligned} \tag{13}$$

Here, $H.c$ is the complex conjugate of $\left\langle\psi\left|U_a^\dagger M\frac{\partial U_a}{\partial\theta_k}\right|\psi\right\rangle$. As the quantum gates in the circuit used in this paper are all Pauli rotation gates,

Table 1
Dataset descriptions.

	Gossip	Politifact
Description	Genuine news data and some instances of false or misleading gossip.	Legitimate and fabricated news data.
Training Set	10 010	381
Real: Fake	2036:7974	135:246
Test Set	1000	100

Table 2
The hyperparameter setting in experiments.

Qubits in QCNN	Learning rate	Batch size	Epoch	Measurement
8	0.001	32	100	Z Gate

according to the conclusions drawn by Schuld et al. [38], there is

$$\begin{aligned}&\left\langle \psi \left| U_a^{\dagger} M \frac{\partial U_a}{\partial \theta_k} \right| \psi \right\rangle + H.c. \\ &= \frac{1}{2}\left(\left\langle \psi \left| U_a^{\dagger}\left(\theta_k + \frac{\pi}{2}\right) M U_a\left(\theta_k + \frac{\pi}{2}\right) \right| \psi \right\rangle \right. \\ &\left. - \left\langle \psi \left| U_a^{\dagger}\left(\theta_k - \frac{\pi}{2}\right) M U_a\left(\theta_k - \frac{\pi}{2}\right) \right| \psi \right\rangle \right)\end{aligned} \tag{14}$$

The method for computing the gradient is called the parameter-shift rule. In this way, the partial derivative of each parameter is obtained and fed into the corresponding optimizer for optimization.

4. Experimental results

4.1. Datasets

This study uses multimodal fake news datasets, namely, Gossip [39] and Politifact [39,40], both collected from the social media platform (Twitter). Each dataset incorporates both text and image. Initially, in each multimodal news, we select its first image (cover image) to participate in the subsequent training. This is because the first image of news usually can represent all the images in this news and carries its most important information. The data in datasets are labeled as real or fake news and divided into training and testing sets, with the validation set extracted directly from the training set. For clarity, a description of two datasets is provided in Table 1.

4.2. Experimental setting

The experimental environment is established on an x86 platform. Data embedding is conducted on a graphics processing unit (NVIDIA RTX3080 10G), and the QCNN training is performed on a central processing unit (Intel Core 12400F).

The hyperparameters used in the experiments are listed in Table 2. Due to the limitation of quantum resources, and considering the effects of the representation capability and the barren plateau phenomenon on the model performance, we use 8 qubits for QCNN representation. Also, the learning rate is set to 0.001, batch size to 32, and epoch to 100. To measure the probability of a quantum state result being 1, the Z measurement operator is employed on the quantum circuit measurements.

When extracting text summaries, the maximum length of each sentence in the news is limited to 30 tokens. A quantum circuit is constructed and trained by using the Pennylane library, with the initial state of the quantum circuit set to $|0\rangle$. Quantum circuit is trained in the environment with simulated quantum noise.

Table 3
The performance comparison of QMFND with other fake news detection models.

Dataset	Model	Accuracy	Recall	Precision	TNR	F1
Gossip	BERT	0.871	0.914	0.924	0.897	0.919
	XLNet	0.884	0.943	0.917	0.897	0.929
	MCAN	0.869	0.806	0.890	0.877	0.846
	MCAN-A	0.851	0.859	0.877	0.859	0.824
	SpotFake	0.789	0.723	0.815	0.788	0.766
	SpotFake+	0.839	0.799	0.853	0.842	0.825
	CNN	0.806	0.643	0.882	0.812	0.744
	QMFND	**0.879**	**0.958**	**0.899**	**0.882**	**0.928**
Politifact	BERT	0.843	0.696	0.904	0.864	0.786
	XLNet	0.847	0.704	0.905	0.867	0.792
	MCAN	0.846	0.829	0.851	0.847	0.84
	MCAN-A	0.809	0.758	0.827	0.81	0.792
	SpotFake	0.779	0.693	0.815	0.775	0.749
	SpotFake+	0.789	0.753	0.803	0.789	0.777
	CNN	0.776	0.614	0.857	0.763	0.715
	QMFND	**0.846**	**0.853**	**0.927**	**0.904**	**0.888**

4.3. Evaluation metrics

Evaluation metrics are adoped as follows: Accuracy, Recall, Precision, True Negative Rate (TNR), and F1, as expressed in Eq. (15). Among them, TP (True Positive) indicates that it correctly predicts the positive class when the sample is actually positive. TN (True Negative) indicates that it correctly predicts the negative class when the sample is actually negative. FP (False Positive) indicates that it incorrectly predicts the positive class when the sample is actually negative. FN (False Negative) indicates that it incorrectly predicts the negative class when the sample is actually positive.

$$\begin{aligned}&Accuracy = \frac{TN + TP}{TP + TN + FP + FN}, Recall = \frac{TP}{TP + FN}, \\ &Precision = \frac{TP}{TP + FP}, \\ &F1 = \frac{2 * Precision * Recall}{Precision + Recall}, TNR = \frac{TN}{TN + FN}.\end{aligned} \tag{15}$$

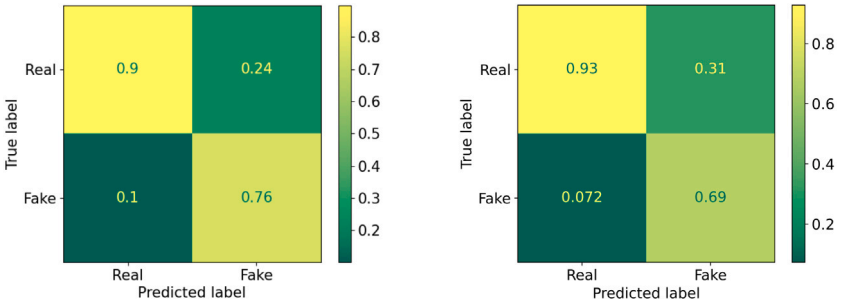
4.4. Baselines

To evaluate the effectiveness of the QMFND model, seven baselines are considered for comparison. These baselines include:

- XLNet: XLNet model [41] is used to verify the authenticity of unimodal (text) data.
- BERT: BERT model [42] to verify the authenticity of unimodal (text) data.
- MCAN: It is a multimodal fake news detection model using a co-attention network, enabling better fusion of textual and visual features for fake news detection [43].
- MCAN-A: It is a model similar to MCAN but without the part of fusing multimodal features. Spatial-domain features, frequency-domain features, and text features are simply connected.
- SpotFake: Multimodal model for detecting fake news without considering additional subtasks such as event discrimination [17].
- SpotFake+: It leverages transfer learning to capture semantic and contextual information from news articles and related images, so as to improve the accuracy of fake news detection [18].
- CNN: It uses the same pre-processing methods as QMFND (XLNet, VGG-19, etc.) and uses a classical CNN for classification tasks.

4.5. Performance analysis

The confusion matrices of the QMFND performance on two datasets are shown in Fig. 10. The comparisons between QMFND and other baselines are presented in Table 3. As shown in the Table 3, QMFND achieves a detection accuracy of 87.9% and 84.6% on the Gossip and Politifact datasets, respectively, which is quite promising. The accuracy



(a) The confusion matrix of QMFND on Gossip. (b) The confusion matrix of QMFND on Politifact.

Fig. 10. The confusion matrices of QMFND on the two datasets.

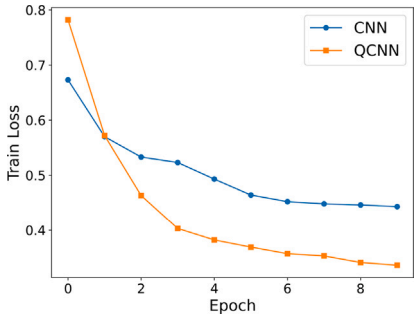


Fig. 11. The comparison of loss function between QCNN and CNN.

of the QMFND model is 0.5% lower than XLNet on the Gossip dataset and 0.1% lower than XLNet on the Politifact dataset. However, on the Politifact dataset, QMFND not only outperforms XLNet in other performance metrics but also outperforms the other six baseline models on both datasets.

In addition, with the integration of a QCNN, QMFND outperforms the structurally equivalent CNN, with a detection accuracy about 7% higher than that of CNN. Moreover, it performs better than most of the baselines. It can be concluded that the improvement in detection performance represents that QCNNs are sharper and more accurate in representing image features. They can capture subtle details that may have been missed by traditional CNNs, thus improving the classification performance.

QCNN is also more efficient. As shown in Fig. 11, the loss function of QCNN decreases faster. This indicates that QMFND utilizes the training data more efficiently within a given number of training iterations, helping to reduce training time and computational cost. Considering the need for efficient processing in big data era, using QCNN implies quicker identification of fake news.

4.6. Parameter effects on the model performance

4.6.1. Effect of different data modalities on detection performance

The performances of unimodal and multimodal on QMFND are compared in this section. As shown in Table 4, on both datasets, QMFND's detection performance is best for text, followed by multimodal, and performs worst on images. In addition, the detection accuracies of text and images in multimodal scenarios are slightly lower than those of unimodal text-only scenarios by 0.5% and 3.8%, respectively, on two datasets.

It is easy to know that, textual information plays a key role in detecting fake news as compared to image information. It can precisely convey the entire content of the news. On the other hand, images alone may not be helpful enough to detect fake news. Although images can present rich information, the key information such as time and location of the events in news, as well as the participants, is usually impossible to be accurately obtained by only images. Therefore, using multimodal data processing methods for detecting fake news becomes more practical and effective by nature.

Table 4
The effect of modal type on detection performance.

DataSet	Modal	Accuracy	Recall	Precision	TNR	F1
Gossip	Image	0.802	0.929	0.842	0.749	0.883
	Text	0.884	0.942	0.917	0.897	0.929
	Multimodal	**0.879**	**0.958**	**0.899**	**0.882**	**0.928**
Politifact	Image	0.692	0.733	0.821	0.719	0.774
	Text	0.884	0.973	0.879	0.882	0.924
	Multimodal	**0.846**	**0.853**	**0.927**	**0.904**	**0.888**

Table 5
The effect of real and fake data proportion on detection performance. "R" represents real news, and "F" represents fake news.

	DataSet	Accuracy	Recall	Precision	TNR	F1
R: F=1:1	Gossip	0.849	0.649	0.940	0.892	0.768
	Politifact	0.840	0.583	0.978	0.932	0.73
R: F=9:1	Gossip	0.879	0.958	0.899	0.882	0.928
	Politifact	0.846	0.853	0.927	0.904	0.888

4.6.2. Effect of data proportion on detection performance

This section explores the effect of the proportion of data labeled as "real" and "fake" in the training dataset on the detection results.

As shown in Table 5, for the real to fake data proportion of 1:1, the accuracies on the Gossip and Politifact datasets are 3% and 0.06% lower compared to those of the 9:1 proportion, respectively. However, in terms of the recall and F1 score, the performance is significantly lower when the real-to-fake data proportion is 1:1 versus 9:1. Specifically, for both datasets, recall is lower by 30.9% and 27%, and the F1 score is lower by 16% and 15.8% for the 1:1 proportion compared to the 9:1 proportion, respectively. This indicates that although the model accuracy on data with a 9:1 real-to-fake proportion does not show an obvious advantage, other performance metrics exhibit clear advantages. This can be explained by the fact that the model focuses more on those classes with a higher percentage of samples during the training process, and learns relatively little about the lower percentage classes. In practical applications, it is necessary to weigh these evaluation metrics according to the specific tasks and goals to deal with the proportion imbalance in datasets.

4.6.3. Effect of the number of qubits and barren plateau phenomenon

The number of qubits represented has an effect on detection performance and the barren plateau phenomenon. With the limited resources available, we compared the performance of QMFND using 4, 8, and 16 qubits for representation. As shown in Table 6, the performance is best when using 8 qubits for representation, followed by 4 qubits, and finally 16 qubits. On both datasets, the accuracy using 8 qubits is 1%

Table 6
The effect of the number of qubits on the detection performance.

DataSet	Qubits	Accuracy	Recall	Precision	TNR	F1
Gossip	4	0.869	0.781	0.9	0.882	0.836
	16	0.827	0.719	0.869	0.835	0.787
	8	**0.879**	**0.958**	**0.899**	**0.882**	**0.928**
Politifact	4	0.836	0.753	0.868	0.844	0.806
	16	0.797	0.71	0.832	0.797	0.766
	8	**0.846**	**0.853**	**0.927**	**0.904**	**0.888**

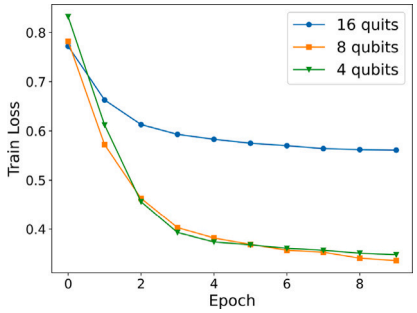


Fig. 12. The loss functions for 4, 8, and 16 qubits quantum circuits.

higher than when using 4 qubits, and with 8 qubits, the recall is higher by 17.7% and 10% compared to that of 4 and 16 qubits, respectively. This indicates that using 8 qubits is the most suitable option for our detection.

The "barren plateau" refers to a phenomenon that the training of QNN becomes inefficient when the number of qubits in quantum computing is large. The objective function tends to flatten, resulting in excessively long training times or training failures [44].

In initial experiments, 16 qubits are used in QCNN. During training, the gradient of the QCNN descends very slowly. As depicted in Fig. 12, a 16-qubit circuit underperforms the 8-qubit or 4-qubit circuit in terms of loss function reduction over the same number of iterations. This is possibly due to the effect of quantum noise. In a limited simulation environment, the noise may make the gradient information lost as the number of qubits increases, making training more difficult. Therefore, to mitigate the effect of the barren plateau phenomenon, 8 qubits are used for representation in QMFND.

4.7. The expressibility and entangling capability of QCNN

The structures of VQCs in QCNNs play a crucial role in the performance of quantum models. Expressibility and entangling capability can be used to assess the quality of VQCs [45].

Expressibility refers to the ability of a circuit to generate states representing the Hilbert space. The quantification of expressibility is defined as the deviation of the distribution of states generated by the set of Haar random states from the states obtained by uniformly sampling the VQC parameters. $\hat{P}_{\mathrm{VQC}}(F;\theta)$ represents the set of states obtained by uniformly sampling parameters. $P_{\mathrm{Haar}}(F)$ represents the set of states obtained from a uniform distribution. $P_{\mathrm{Haar}}(F) = (N-1)(1-F)^{N-2}$, where F corresponds to fidelity, and N is the dimension of the Hilbert space. D_{KL} is for calculating the KL divergence between two sets [46]. Then, the expressibility of a VQC can be expressed as

$$\mathrm{Expr} = D_{\mathrm{KL}}\left(\hat{P}_{\mathrm{VQC}}(F;\theta) \parallel P_{\mathrm{Haar}}(F)\right). \tag{16}$$

In VQCs, a higher degree of entanglement in the circuit implies a better representation of the solution spaces for tasks such as data classification, as well as better capture of quantum data correlations. The Meyer–Wallach (MW) measurement is used to quantify the entangling capability of the VQC. For a given VQC, this value is estimated by sampling the circuit parameters and calculating the sample average

Table 7
The expressibility and entangling capability of the designed QCNN circuit.

	Conv1	Conv2	Pool1	Pool2	QCNN
Entangling capability	0.498	0.497	0.5	0.25	**0.819**
Expressibility	2.311	1.472	2.31	1.479	**0.00015**

of the MW measurements of the output states. Here, θ represents the circuit parameters. $|\vec{\theta}|$ is the number of test parameters, n is the number of parameters, and ρ_k is the partial trace. The MW measurement formulation is expressed as follows:

$$Q^{MW} = \frac{2}{|\vec{\theta}|}\sum_{\theta_i\in\vec{\theta}}\left(1-\frac{1}{n}\sum_{k=1}^{n}\mathrm{Tr}\left(\rho_k^2\left(\theta_i\right)\right)\right). \tag{17}$$

The expressibility and entangling capability of the designed QCNN circuit are listed in Table 7. The entangling capability of quantum circuits is from 0 to 1, with the higher value being preferable. On the contrary, for quantum circuits, lower values of expressibility are desirable. It is evident that the VQC of QCNN in this paper exhibits excellent expressibility and entangling capability.

4.8. Robustness analysis against quantum noise

In the noisy intermediate-scale quantum (NISQ) era, the implications of quantum noise must be considered when developing QML models. Noise originating from the quantum channels frequently disrupts the quantum entanglement state, which can adversely affect the performance of QML models. Quantum noise not only undermines model precision but can potentially lead to a barren plateau phenomenon. Quantum noise associated with a single qubit can be described using the Kraus matrices [47]. The Kraus matrices of bit flip (BF) are

$$K_0 = \sqrt{1-p}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, K_1 = \sqrt{p}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} \tag{18}$$

The Kraus matrices of phase flip (PF) noise are

$$K_0 = \sqrt{1-p}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, K_1 = \sqrt{p}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} \tag{19}$$

The Kraus matrices of amplitude damping (AD) noise are

$$K_0 = \begin{bmatrix}1 & 0\\ 0 & \sqrt{1-p}\end{bmatrix}, K_1 = \begin{bmatrix}0 & \sqrt{p}\\ 0 & 0\end{bmatrix} \tag{20}$$

The Kraus matrices of depolarization noise (DN) noise are

$$K_2 = \sqrt{p/3}\begin{bmatrix}0 & -i\\ i & 1\end{bmatrix}, K_3 = \sqrt{p/3}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} \tag{21}$$

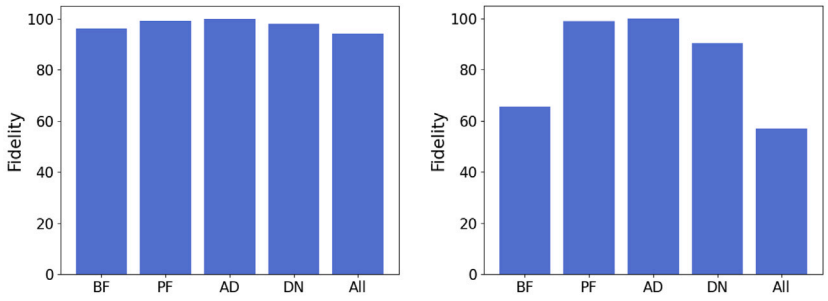
where p denotes the probability of noise occuring on a single qubit. $p \in \{0,1\}$.

The robustness of quantum algorithms against noise is assessed using the fidelity metric. Fidelity serves as a measure of the closeness between two quantum states, representing the probability of one quantum state being recognized as another after testing. A high-quality quantum algorithm possesses high fidelity. Thus, fidelity can be perceived as an indicator of the extent of the effect of quantum noise on algorithmic outcomes. The lower the effect of noise on the results is, the higher its fidelity will be. For two mixed states, the formulation for fidelity is expressed as follows:

$$F(\rho,\sigma) = Tr\left(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right)^2. \tag{22}$$

where ρ and σ are the density matrices of the two mixed states.

In this study, the fidelities of the QCNN circuit are calculated in four single noise channels and a noise combination channel of all noise types with $p = 0.1$ and $p = 0.01$ respectively. "All" represents the noise



(a) The fidelities in noise channels with $p = 0.01$. (b) The fidelities in noise channels with $p = 0.1$.

Fig. 13. The fidelities of the QCNN circuit in noise channels with different p values.

Table 8
The fidelities of our QCNN circuit in different noise channels with $p = 0.1$ and $p = 0.01$.

Noise type	$p = 0.01$	$p = 0.1$
BF	96.06	65.6
PF	99.12	99.01
AD	99.99	99.99
DN	98.01	90.37
All	94.2	57.03

Table 9
The comparison of time complexity between QMFND and the baselines.

Model	Time complexity
BERT [42]	$O(n)$
XLNet [41]	$O(n)$
MCAN/MCAN-A [43]	$O(n)$
SpotFake [17]/SpotFake+ [18]	$O(n)$
CNN	$O(n)$
QMFND	$O\left(\log_2 n\right)$

combination channel by the superposition of all noise operators (Kraus matrices) on single qubits. The results are shown in Table 8 and Fig. 13.

It is not difficult to find that the BF noise has a greater impact on the fidelity of QCNN. In QCNNs, qubits are usually used to represent the features of the input data. If a qubit flips, the feature information originally encoded can be lost. In contrast, other noises may affect the phase or amplitude of the qubits without directly causing the loss of information. Also, when the noise increases, the effect of the BF noise will be obvious. This is possible because, in a QCNN, information will pass through different layers. If a qubit flips, its information in the subsequent layers will also be affected, which can lead to cumulative errors throughout the network. Although the QCNN circuit in this study is somewhat affected by the BF noise when $p = 0.1$, it consistently upholds high fidelity. Notably, for both PF and AD noises, the fidelities are nearly 1, suggesting a minimal influence of these noise types on performance. This attests to the robustness of the QCNN against quantum noise.

4.9. Complexity analysis

The complexity mainly includes quantum circuit complexity and time complexity. Quantum circuit complexity can be calculated by the number of quantum gates used in the QCNN. As depicted in Figs. 7 and 8, eight quantum gates are required in a quantum convolutional layer, and six quantum gates are required in the quantum pooling layer. Suppose N qubits satisfy $N = 2^a$ (a is an integer greater than or equal to 1), then, the number of quantum gates required in the $i-$th layer is $14N/2^i$, where $i \leqslant a$. And the total number of quantum gates required by the QMFND model is calculated as:

$$2 \cdot \left(\frac{N}{2} \cdot 7 + \frac{N}{2^2} \cdot 7 + \cdots + \frac{N}{2^{a-1}} \cdot 7\right) + 14 = 7N\left(2 - \frac{1}{2^{a-2}}\right) + 14 = 14\left(N - 1\right). \tag{23}$$

Therefore, the total circuit complexity is $O(N)$.

A lower time complexity in a CNN or QCNN model typically indicates higher computational efficiency of its network. In other words, for a given task, the model's computational cost is relatively low. Comparing the QCNN in QMFND with that in baselines, as shown in Table 9, the complexity of executing a CNN to generate n-dimensional classical data is $O(n)$. However, for a QCNN using quantum amplitude encoding, only $\lceil \log_2 n \rceil$ qubits are required to represent an n-dimensional vector. Therefore, the time complexity of QCNN models can be minimized as $O\left(\log_2 n\right)$. This emphasizes the benefits of employing qubits for feature representation. They have the capability to capture a greater number of features while utilizing fewer resources compared to classical computer data, which is more efficient.

5. Conclusion

For quicker and more accurate detection of fake news on social media, the QMFND model is proposed by combining quantum perspectives. The experimental results show that QMFND achieves high accuracies of 87.9% and 84.6% on two datasets. QMFND outperforms the six baselines and has only a slight disadvantage compared to XLNet. The designed QCNN circuit has good expressibility, entangling capability, and robustness against quantum noise. The complexity of QMFND is much lower than the classical models by combining amplitude encoding to encode multimodal data into quantum states. However, the performance of QMFND can be compromised by current hardware limitations and substantial BF noise in the running environment of quantum computers. In future, We can introduce quantum error-correcting codes like Shor codes to minimize the effect of noise. Also, to improve the accuracy, we can consider quantum fuzzy neural networks to increase the network's ability to adapt to uncertainty. These considerations will help to improve the performance of fake information detection and develop more efficient and reliable quantum-based models.

CRediT authorship contribution statement

Zhiguo Qu: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft. **Yunyi Meng:** Data curation, Formal analysis, Methodology, Supervision, Writing – original draft. **Ghulam Muhammad:** Funding acquisition, Investigation, Project administration, Supervision, Writing – review & editing. **Prayag Tiwari:** Data curation, Formal analysis, Investigation, Project administration, Software, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

https://github.com/olivia-2333/QMFND_quantum_fake_news_detection.

Acknowledgment

This work was supported by the Deputyship for Research and Innovation, "Ministry of Education," Saudi Arabia, under Grant IFKSUOR3-283-2.

References

[1] E.C. Tandoc Jr., The facts of fake news: A research review, Sociol. Compass 13 (9) (2019) e12724.
[2] T. Meng, X. Jing, Z. Yan, W. Pedrycz, A survey on machine learning for data fusion, Inf. Fusion 57 (2020) 115–129.
[3] L. Wei-bin, Z. Zhi-yuan, X. Wei-wei, Feature fusion methods in pattern classification, J. Beijing Univ. Posts Telecommun. 40 (4) (2017) 1.
[4] U.G. Mangai, S. Samanta, S. Das, P.R. Chowdhury, A survey of decision fusion and feature fusion strategies for pattern classification, IETE Tech. Rev. 27 (4) (2010) 293–307.
[5] J. Yao, V.V. Raghavan, Z. Wu, Web information fusion: A review of the state of the art, Inf. Fusion 9 (4) (2008) 446–449.
[6] H.T. Phan, N.T. Nguyen, D. Hwang, Fake news detection: A survey of graph neural network methods, Appl. Soft Comput. (2023) 110235.
[7] N.R. de Oliveira, P.S. Pisa, M.A. Lopez, D.S.V. de Medeiros, D.M. Mattos, Identifying fake news on social networks based on natural language processing: Trends and challenges, Information 12 (1) (2021) 38.
[8] R. Oshikawa, J. Qian, W.Y. Wang, A survey on natural language processing for fake news detection, 2018, arXiv preprint arXiv:1811.00770.
[9] Y. Wang, F. Ma, Z. Jin, Y. Yuan, G. Xun, K. Jha, L. Su, J. Gao, Eann: Event adversarial neural networks for multi-modal fake news detection, in: Proceedings of the 24th Acm Sigkdd International Conference on Knowledge Discovery & Data Mining, 2018, pp. 849–857.
[10] G.L. Long, L. Xiao, Parallel quantum computing in a single ensemble quantum computer, Phys. Rev. A 69 (5) (2004) 052303.
[11] M. Sasaki, A. Carlini, Quantum learning and universal quantum matching machine, Phys. Rev. A 66 (2) (2002) 022303.
[12] H. Neven, V.S. Denchev, G. Rose, W.G. Macready, Training a large scale classifier with the quantum adiabatic algorithm, 2009, arXiv preprint arXiv:0912.0779.
[13] D. Anguita, S. Ridella, F. Rivieccio, R. Zunino, Quantum optimization for training support vector machines, Neural Netw. 16 (5–6) (2003) 763–770.
[14] E. Ovalle-Magallanes, J.G. Avina-Cervantes, I. Cruz-Aceves, J. Ruiz-Pinales, Hybrid classical–quantum convolutional neural network for stenosis detection in X-ray coronary angiography, Expert Syst. Appl. 189 (2022) 116112.
[15] Z. Qu, X. Liu, M. Zheng, Temporal-spatial quantum graph convolutional neural network based on Schrödinger approach for traffic congestion prediction, IEEE Trans. Intell. Transp. Syst. (2022).
[16] J. Ma, W. Gao, K.-F. Wong, Detect rumors on twitter by promoting information campaigns with generative adversarial learning, in: The World Wide Web Conference, 2019, pp. 3049–3055.
[17] S. Singhal, R.R. Shah, T. Chakraborty, P. Kumaraguru, S. Satoh, Spotfake: A multi-modal framework for fake news detection, in: 2019 IEEE Fifth International Conference on Multimedia Big Data, BigMM, IEEE, 2019, pp. 39–47.
[18] S. Singhal, A. Kabra, M. Sharma, R.R. Shah, T. Chakraborty, P. Kumaraguru, Spotfake+: A multimodal framework for fake news detection via transfer learning (student abstract), in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, (no. 10) 2020, pp. 13915–13916.
[19] I. Segura-Bedmar, S. Alonso-Bartolome, Multimodal fake news detection, Information 13 (6) (2022) 284.
[20] N. Jayakody, A. Mohammad, M.N. Halgamuge, Fake news detection using a decentralized deep learning model and federated learning, in: IECON 2022–48th Annual Conference of the IEEE Industrial Electronics Society, IEEE, 2022, pp. 1–6.
[21] A.M. Luvembe, W. Li, S. Li, F. Liu, G. Xu, Dual emotion based fake news detection: A deep attention-weight update approach, Inf. Process. Manage. 60 (4) (2023) 103354.
[22] V.H. Nguyen, K. Sugiyama, P. Nakov, M.Y. Kan, FANG: Leveraging social context for fake news detection using graph representation, 2020.
[23] S. Raza, C. Ding, Fake news detection based on news content and social contexts: A transformer-based approach, Int. J. Data Sci. Anal. 13 (4) (2022) 335–362.
[24] J. Li, S. Ni, H.-Y. Kao, Meet the truth: Leverage objective facts and subjective views for interpretable rumor detection, 2021, arXiv preprint arXiv:2107.10747.
[25] L. Hu, T. Yang, L. Zhang, W. Zhong, D. Tang, C. Shi, N. Duan, M. Zhou, Compare to the knowledge: Graph neural fake news detection with external knowledge, in: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), 2021, pp. 754–763.
[26] B. Narottama, S.Y. Shin, Quantum neural networks for resource allocation in wireless communications, IEEE Trans. Wireless Commun. 21 (2) (2021) 1103–1116.
[27] P. Tiwari, L. Zhang, Z. Qu, G. Muhammad, Quantum fuzzy neural network for multimodal sentiment and sarcasm detection, Inf. Fusion (2023) 102085.
[28] V. Rajesh, U.P. Naik, et al., Quantum convolutional neural networks (QCNN) using deep learning for computer vision applications, in: 2021 International Conference on Recent Trends on Electronics, Information, Communication & Technology, RTEICT, IEEE, 2021, pp. 728–734.
[29] Y. Li, R.-G. Zhou, R. Xu, J. Luo, W. Hu, A quantum deep convolutional neural network for image recognition, Quantum Sci. Technol. 5 (4) (2020) 044003.
[30] G. Chen, Q. Chen, S. Long, W. Zhu, Z. Yuan, Y. Wu, Quantum convolutional neural network for image classification, Pattern Anal. Appl. 26 (2) (2023) 655–667.
[31] C.-H.H. Yang, J. Qi, S.Y.-C. Chen, P.-Y. Chen, S.M. Siniscalchi, X. Ma, C.-H. Lee, Decentralizing feature extraction with quantum convolutional neural network for automatic speech recognition, in: ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2021, pp. 6523–6527.
[32] Z. Qu, Y. Li, P. Tiwari, QNMF: A quantum neural network based multimodal fusion system for intelligent diagnosis, Inf. Fusion 100 (2023) 101913.
[33] Z. Qu, W. Shi, B. Liu, D. Gupta, P. Tiwari, IoMT-based smart healthcare detection system driven by quantum blockchain and quantum neural network, IEEE J. Biomed. Health Inform. (2023).
[34] I. Cong, S. Choi, M.D. Lukin, Quantum convolutional neural networks, Nat. Phys. 15 (12) (2019) 1273–1278.
[35] W. Jiang, J. Xiong, Y. Shi, A co-design framework of neural networks and quantum circuits towards quantum advantage, Nat. Commun. 12 (1) (2021) 579.
[36] D. Kartsaklis, I. Fan, R. Yeung, A. Pearson, R. Lorenz, A. Toumi, G. de Felice, K. Meichanetzidis, S. Clark, B. Coecke, Lambeq: An efficient high-level python library for quantum NLP, 2021, arXiv preprint arXiv:2110.04236.
[37] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, 2014, arXiv preprint arXiv:1409.1556.
[38] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, N. Killoran, Evaluating analytic gradients on quantum hardware, Phys. Rev. A 99 (3) (2019) 032331.
[39] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, H. Liu, FakeNewsNet: A data repository with news content, social context and dynamic information for studying fake news on social media, 2018, arXiv preprint arXiv:1809.01286.
[40] Tampa Bay Times, Politifact, 2007, https://github.com/KaiDMML/FakeNewsNet. (Accessed 2007).
[41] Z. Yang, Z. Dai, Y. Yang, J. Carbonell, R.R. Salakhutdinov, Q.V. Le, Xlnet: Generalized autoregressive pretraining for language understanding, Adv. Neural Inf. Process. Syst. 32 (2019).
[42] I. Turc, M.-W. Chang, K. Lee, K. Toutanova, Well-read students learn better: On the importance of pre-training compact models, 2019, arXiv preprint arXiv:1908.08962v2.
[43] Y. Wu, P. Zhan, Y. Zhang, L. Wang, Z. Xu, Multimodal fusion with co-attention networks for fake news detection, in: Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021, 2021, pp. 2560–2569.
[44] J.R. McClean, S. Boixo, V.N. Smelyanskiy, R. Babbush, H. Neven, Barren plateaus in quantum neural network training landscapes, Nat. Commun. 9 (1) (2018) 4812.
[45] S. Sim, P.D. Johnson, A. Aspuru-Guzik, Expressibility and entangling capability of parameterized quantum circuits for hybrid quantum-classical algorithms, Adv. Quantum Technol. 2 (12) (2019).
[46] K. yczkowski, H.-J. Sommers, Average fidelity between random quantum states, Phys. Rev. A 71 (3) (2005) 32313.
[47] M.A. Nielsen, I. Chuang, Quantum Computation and Quantum Information, American Association of Physics Teachers, 2002.